Every time you need to describe a data structure, you can follow a few simple steps to make sure to provide a comprehensive and unambiguous specification.

- Specifying its API first, with a focus on the methods' input and output;
- Describing its high-level behavior;
- Describing in detail the behavior of its concrete implementation;
- Analyzing the performance of its methods.

We will use the same workflow for the data structures presented in this book, after describing a concrete scenario in which each data structure is actually used.

Starting in chapter 2, with the description of the first data structure presented, we will also explain in further detail the conventions we use for the API description.

1.3 Setting Goals: Your Expectations After Reading this Book

One question you might have, by now, is: “will I ever need to write my own data structures?”.

There is a good chance that you will rarely find yourself in a situation where you don’t have any alternative to writing a data structure from scratch; today it isn’t difficult to find libraries implementing the most common data structures in